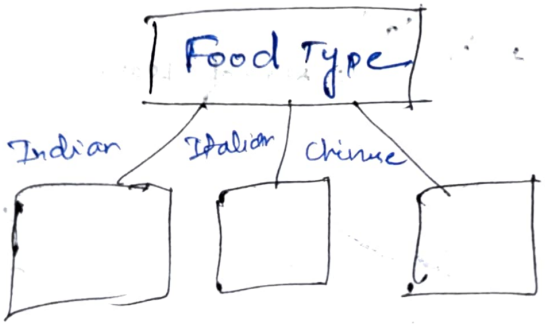
16/02/2026

1. Information Gain
2. Gini Index
3. Binary Cross Entropy

classification Tree



weather	windy	Play
Sunny	N	N
Sunny	Y	N
overcast	N	Y
Rainy	N	Y
Rainy	Y	N
Rainy	N	Y
Overcast	Y	Y
Sunny	N	N

Step: 1 : compute root entropy

Yes = 4 , NO = 4

$P(Yes) = \frac{4}{8} = 0.5 \quad P(NO) = \frac{4}{8} = 0.5$

Entropy, H

$H_{root} = \sum -P_i \log_2 P_i$

$= -P_{yes} \log_2 P_{yes} + -P_{NO} \log_2 P_{NO}$

$= -0.5 \log_2(0.5) + -0.5 \log_2(0.5)$

$H_{root} = 1$

Step-2: compute information gain (IG) for all feature = Weather

Sunny

3 sample S_1, S_2, S_8 (NO, NO, NO) $P_{NO} = 3/3 = 1$

$$H_{sunny} = \sum_{i=1}^{1} -P_i \log_2 P_i$$

$$= -1 \log_2 1$$

$$= 0$$ ⇒ Entropy is zero which means zero impurity is pure

$$H_{overcast} = \sum_{i=1}^{1} -P_i \log_2 P_i$$

$$= -1 \log_2 1$$

$$= 0$$

H_{Rain} ⇒ 3 sample (Yes, NO, Yes) $P_{Yes} = 2/3$ $P_{NO} = 1/3$

$$= \sum_{i=1}^{2} -2/3 \log_2 2/3 + -1/3 \log_2 1/3$$

$$= -\frac{2}{3}(-0.585) + -1/3(-1.585)$$

$$= 0.39 + 0.52$$

$$= 0.91$$

compute weight entropy

$$H_{weight} = \frac{\text{No. of sample}}{\text{total sample}} H_{child}$$

$$= \frac{3}{8} * 0 + \frac{2}{8} * 0 + \frac{3}{8} * 0.912$$

$$= 0.344$$

$$IG_{weather} = H_{parent} - H_{child\ (weighted)}$$

$$= 1 - 0.344$$

$$= 0.656$$

arg max (IG)

Pick the feature that give max IG value